

ngspice commands — a worked reference

Every command in this ngspice-46 + OpenVAF-r build, with netlists and plots

Ngspice + OpenVAF Enhancements project

July 2026

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ngspice commands — a worked reference

This document walks through the commands available in this ngspice build — the stock nutmeg/spice command set plus every command added by the [enhancement project](#) (the RF steady-state suite, the optimizer, Monte-Carlo/yield, post-layout reliability, pyplot, pre_osdi/pre_snp, eye, stb, ...). The core analyses are shown with complete, copy-pasteable netlists and plots produced by actually running them; the many smaller utility commands are collected into reference tables at the end.

Every plot in this document was generated by [make_commands_figs.py](#) from the exact netlists shown.

1. How commands are run

An ngspice input file (a *deck* or *netlist*) has this shape — the first line is always the title and is ignored as a component:

```
* my circuit title          <- title line (ignored)
V1 in 0 dc 1                <- components
R1 in out 1k
C1 out 0 1u
.tran 10u 5m                <- a "dot-card" analysis
.control                    <- an interactive command block
run
```

```
plot v(out)
.endc
.end
```

There are four ways a command runs:

Context	How	Example
Batch	ngspice -b deck.cir runs every dot-card analysis and any .control block, then exits	ngspice -b amp.cir
Interactive	at the ngspice N -> prompt	ngspice -> tran 1u 1m run then print v(out)
.control block	commands between .control and .endc in the deck	.ac dec 10 1 1meg
Dot-card	.tran/.ac/... in the deck body, executed by run	

The interactive analysis commands (tran, ac, dc, op, ...) take the same arguments as their dot-card spelling and run immediately. run executes the analysis named by the deck's dot-cards. Results are stored as named vectors inside plots (print, plot, and expressions read them back).

2. Analyses

op — DC operating point

Solves the quiescent (DC) bias point: every node voltage and source current with all capacitors open and inductors shorted.

```
* operating point of a resistive divider
V1 in 0 dc 5
R1 in out 2k
R2 out 0 3k
.control
op
print v(out) i(v1)
.endc
.end
```

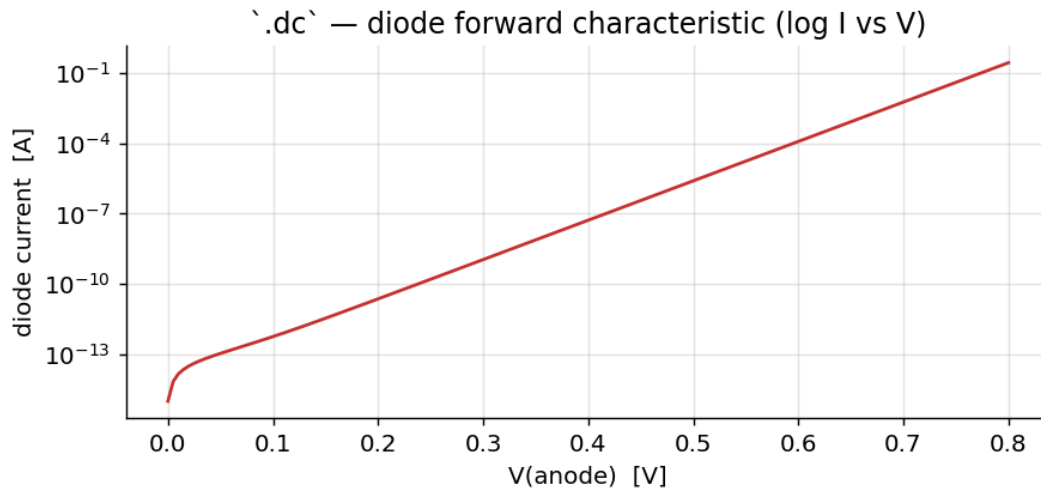
v(out) = 3.0 V, i(v1) = -1 mA. op is also the bias point every AC, noise, tf, pz, and small-signal analysis linearizes around.

dc — DC sweep (and nested sweep)

dc <src> <start> <stop> <step> [<src2> <start2> <stop2> <step2>] sweeps a source (and optionally a second, giving a family of curves). The classic uses are a diode I-V and a transistor's output characteristics.

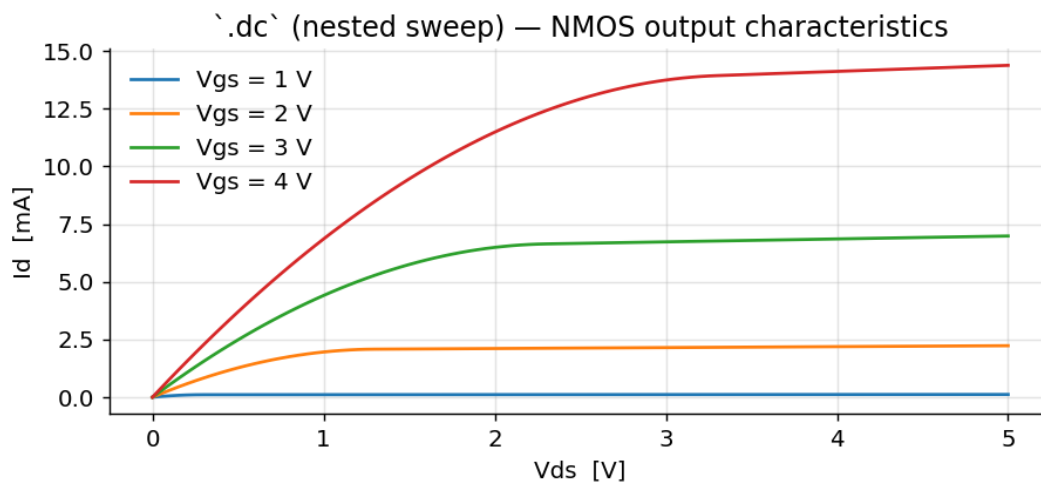
```
* diode forward I-V
Vd a 0 dc 0
D1 a 0 DMOD
.model DMOD D(is=1e-14 n=1)
.dc Vd 0 0.8 0.005
.control
run
```

```
plot abs(i(vd)) ylog
.endc
.end
```



A nested sweep produces the textbook MOSFET output family — the inner source V_{ds} sweeps for each value of the outer source V_{gs} :

```
* NMOS output characteristics (nested .dc)
Vds d 0 dc 0
Vgs g 0 dc 0
M1 d g 0 0 NM W=20u L=1u
.model NM NMOS(level=1 vto=0.7 kp=120u lambda=0.02)
.dc Vds 0 5 0.05 Vgs 1 4 1
.control
run
plot -i(vds)
.endc
.end
```



ac — small-signal frequency response

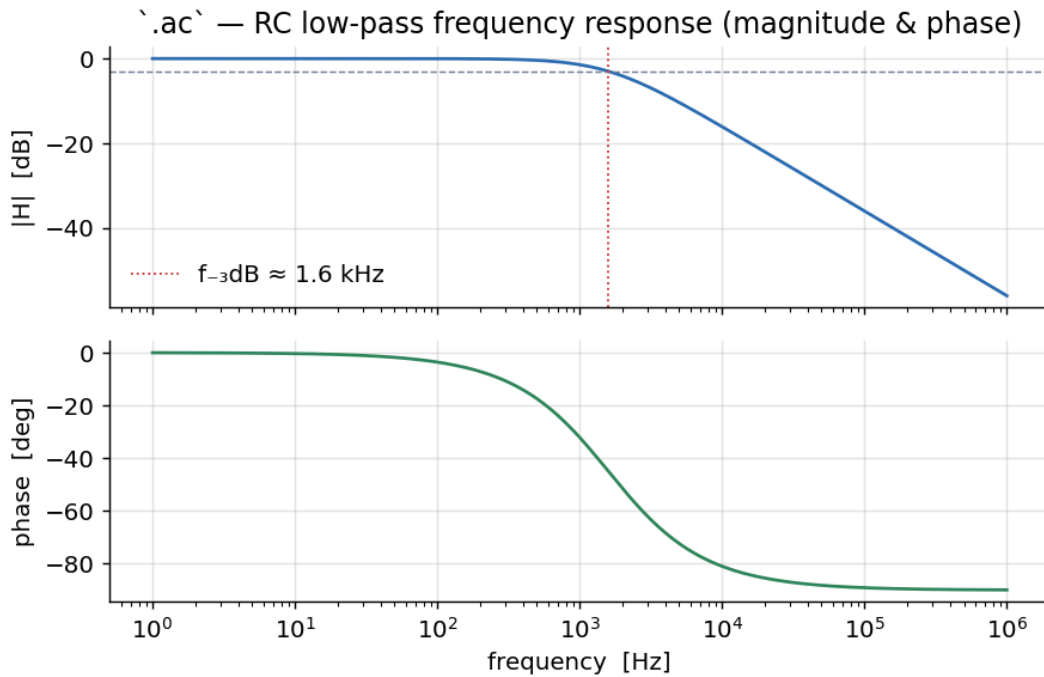
`ac <lin|dec|oct> <N> <fstart> <fstop>` linearizes at the operating point and sweeps frequency. Drive the input with an ac magnitude and read `vdb()` (magnitude in dB) and `vp()` (phase).

* RC low-pass Bode

```

Vin in 0 dc 0 ac 1
R1 in out 1k
C1 out 0 100n
.ac dec 40 1 1meg
.control
run
plot vdb(out)
plot vp(out)
.endc
.end

```



The -3 dB corner sits at $1/(2\pi RC) \approx 1.6$ kHz; the phase rolls from 0° to -90° .

tran — transient (time-domain)

`tran <tstep> <tstop> [tstart [tmax]] [uic]` integrates the circuit in time.

* 1 V step into an RC (over-damped) and a series RLC (ringing)

```
Vin in 0 pulse(0 1 1u 1n 1n 1 2)
```

```
R1 in a 1k
```

```
C1 a 0 100n
```

```
L1 in b 1m
```

```
Rs b c 20
```

```
C2 c 0 100n
```

```
.tran 1u 400u
```

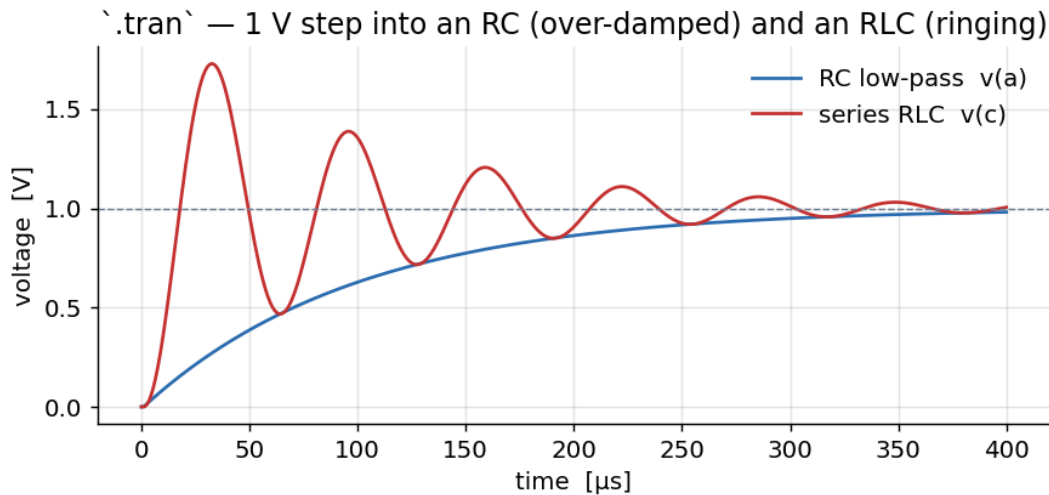
```
.control
```

```
run
```

```
plot v(a) v(c)
```

```
.endc
```

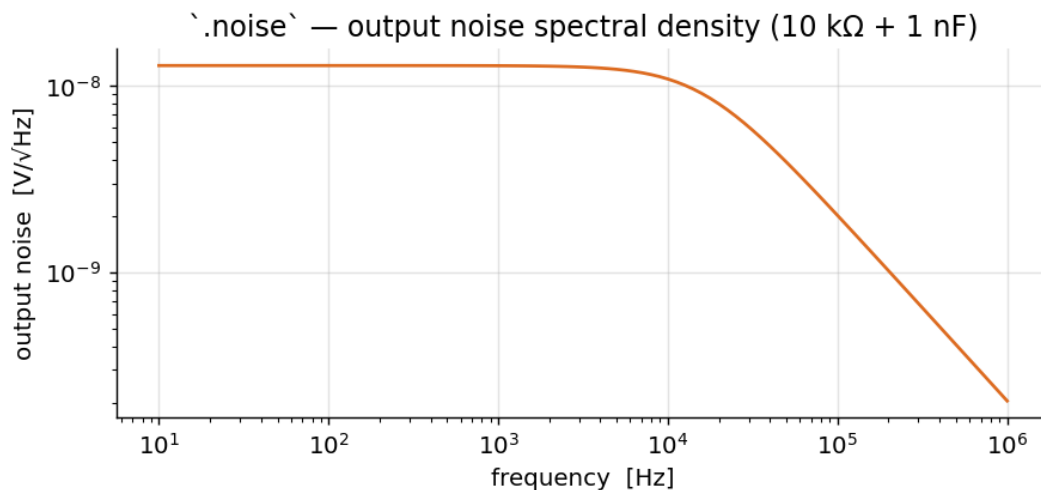
```
.end
```



noise — noise analysis

`noise <output> <input-source> <lin|dec|oct> <N> <fstart> <fstop>` computes the output- and input-referred noise spectral density (and integrated noise). The spectrum vectors live in the `noise1` plot; the integrated totals in `noise2`.

```
* output noise of a 10k + 1n RC
Vin in 0 dc 0 ac 1
R1 in out 10k
C1 out 0 1n
.noise v(out) Vin dec 20 10 1meg
.control
run
setplot noise1
plot onoise_spectrum
.endc
.end
```



tf — DC transfer function

`tf <output> <input-source>` reports the small-signal DC gain plus the input and output impedance. `<output>` is `v(node)`, `v(node1,node2)`, or `i(vsource)`.

```
* transfer function of a voltage divider
```

```

Vin in 0 dc 1
R1 in out 1k
R2 out 0 1k
.control
tf v(out) Vin
.endc
.end

```

Reports transfer_function = 0.5, output_impedance_at_v(out) = 500 Ω , vin#input_impedance = 2 k Ω .

pz — pole/zero

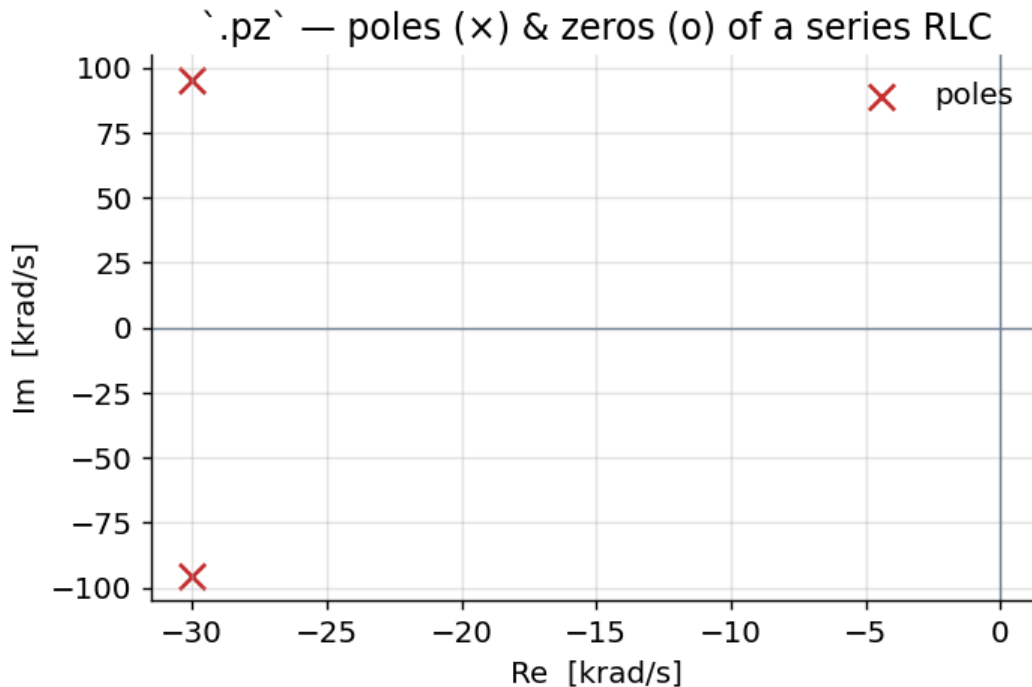
pz <n1> <n2> <n3> <n4> <cur|vol> <pz|pol|zer> finds the poles and zeros of the transfer function between the input node pair (n1,n2) and output pair (n3,n4).

* poles/zeros of a series RLC (v(out)/vin)

```

Vin in 0 dc 0 ac 1
L1 in m 1m
R1 m out 60
C1 out 0 100n
.pz in 0 out 0 vol pz
.control
run
print all
.endc
.end

```



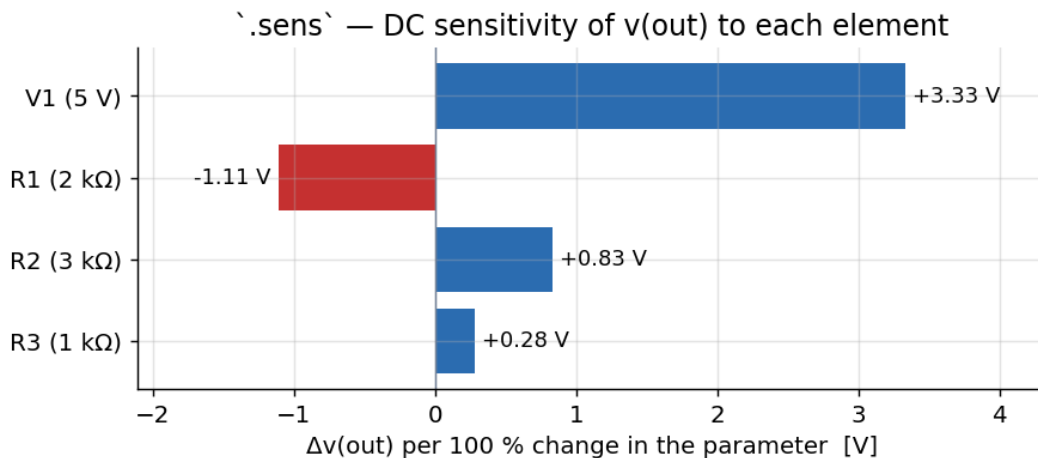
The complex-conjugate pole pair sits at -30 ± 95 krad/s — a resonance near $\omega_0 = 1/\sqrt{LC} = 100$ krad/s with the loss set by R.

sens — sensitivity

sens <output> computes the DC sensitivity of an output to every circuit parameter; append **ac** <lin|dec|oct> <N> <fstart> <fstopt> for the small-signal AC sensitivity vs frequency. The result lands in a **sens1** plot, one vector per device (its value sensitivity) and per model parameter (<dev>: <param>).

```
* DC sensitivity of v(out) to every element
V1 in 0 dc 5
R1 in out 2k
R2 out mid 3k
R3 mid 0 1k
.sens v(out)
.control
run
print v1 r1 r2 r3      ; each is  $\partial v(\text{out})/\partial \text{param}$ 
.endc
.end
```

Scaling each raw sensitivity by the parameter's nominal value gives a comparable “ $\Delta v(\text{out})$ per 100 % change” — the clearest way to rank which element matters most:



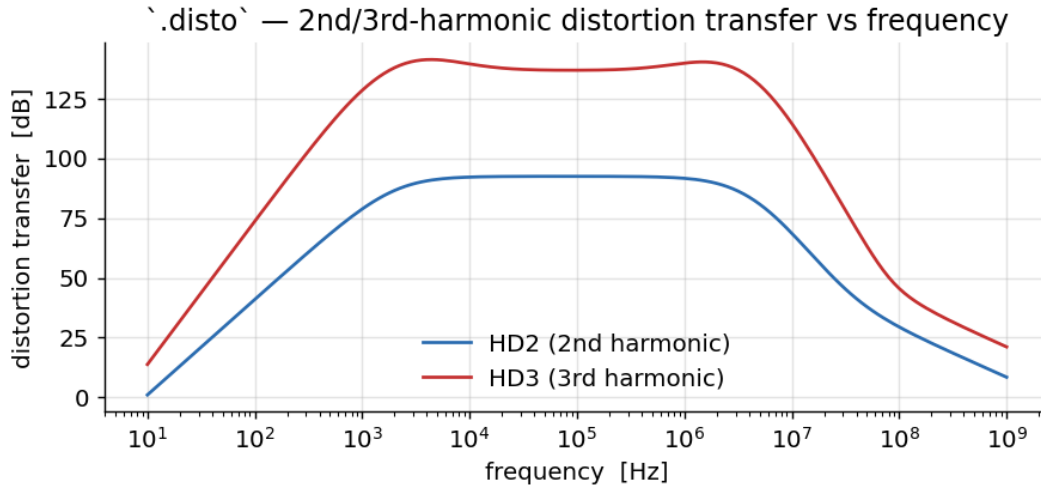
The output tracks the 5 V supply most strongly, and increasing the series R_1 *lowers* it (negative sensitivity), as expected for a divider.

disto — small-signal distortion

disto <lin|dec|oct> <N> <fstart> <fstopt> [f2overf1] sweeps a small-signal harmonic (and, with a second tone, intermodulation) distortion analysis. Mark the driving source with **distof1** (and **distof2** for two-tone). The 2nd- and 3rd-harmonic products land in the **disto1** / **disto2** plots.

```
* small-signal distortion of a common-emitter BJT amp
Vcc cc 0 dc 5
Vin in 0 dc 0.75 ac 1 distof1 1
Rb cc b 220k
Rc cc c 2.2k
Cin in b 1u
Cout c 0 20p
Q1 c b 0 QMOD
.model QMOD NPN(is=1e-15 bf=150 vaf=80 cjc=3p tf=0.3n)
.disto dec 20 10 1g
```

```
.control
run
setplot disto1
plot vdb(c)          ; HD2 vs frequency (disto2 holds HD3)
.endc
.end
```



The distortion (here in dB relative to the unit `distof1` drive) is flat across the amplifier's passband and rolls off with the input coupling capacitor at low frequency and the transistor's bandwidth at high frequency.

fourier

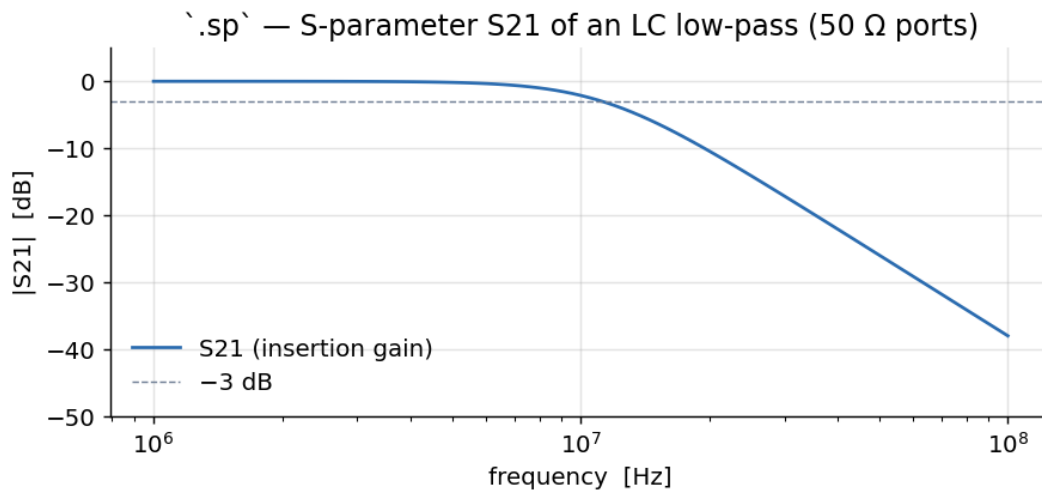
`fourier <fund_freq> <vec>` prints the DC term, the first harmonics, and the THD of a time-domain waveform — run a `.tran` first, then `fourier 10k v(out)`. (The `spec/fft` commands in §4 give the full spectrum as a plot.)

3. RF and periodic steady-state

The project adds a full RF suite on top of the analyses above. Most run as their own command (or dot-card); the periodic small-signal analyses (`.pac`, `.pnoise`, `.pxf`, `.psp`) run as dot-cards around a `pss/.pss` operating point.

`sp` is the small-signal S-parameter analysis — tag the ports with `portnum/z0` sources:

```
* LC low-pass S-parameters (50 ohm ports)
L1 in out 1u
C1 out 0 400p
V1 in 0 dc 0 ac 1 portnum 1 z0 50
V2 out 0 dc 0 ac 1 portnum 2 z0 50
.sp dec 50 1meg 100meg
.control
run
plot vdb(S_2_1)
.endc
.end
```

Command	What it does
<code>sp</code>	S-parameters + noise figure (<code>.sp</code>), Touchstone-exportable
<code>pss</code>	periodic steady state (shooting); the base for the periodic small-signal set
<code>hb</code>	harmonic-balance steady-state spectrum: <code>hb <f0> <K></code>
<code>hbosc</code>	autonomous HB for an oscillator: <code>hbosc <oscnode> <K> [fguess] [tstab]</code>
<code>phasenoise</code>	oscillator phase-noise $L(\Delta f)$ around an <code>hbosc</code> point
<code>qpss</code>	two-tone quasi-periodic steady state: <code>qpss <expr> <f1> <f2> ...</code>
<code>qpac /</code> <code>qpxf /</code> <code>qpnoise</code>	two-tone periodic AC / transfer / noise around a <code>qpss...hb</code> point
<code>envelope</code>	envelope following: <code>envelope <node> <fc> <tstop></code>
<code>stb</code>	loop-gain stability (Middlebrook/Tian double injection); reports phase/gain margin
<code>loadpull</code>	PA load-pull (and source-pull, <code>-source</code>): sweep the load impedance Γ over the Smith chart, per-point large-signal <code>.tran</code> , contour Pout / gain / PAE / efficiency; view with <code>pyplot -contour gamma_re gamma_im pout_dbm</code>

Periodic small-signal — `.pss`, `.pac`, `.pnoise`, `.pxf`, `.psp`

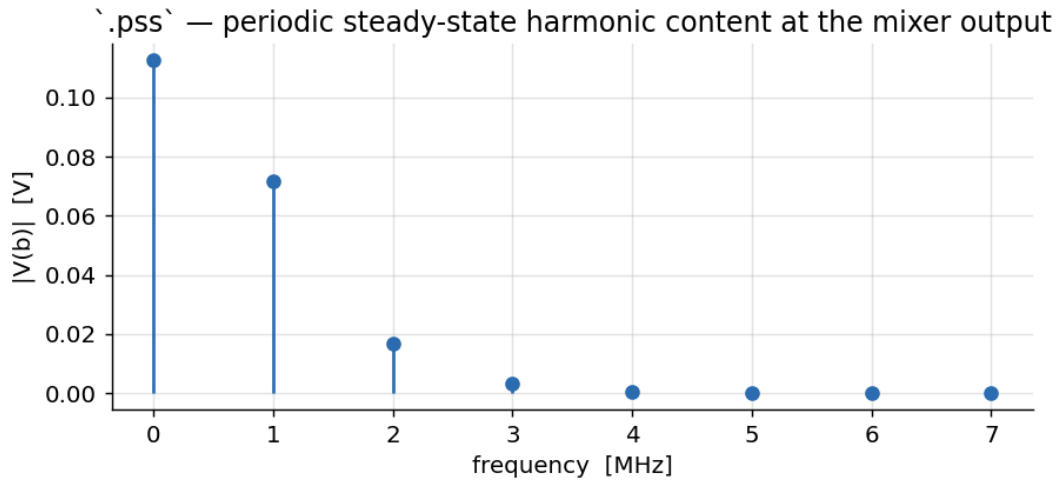
These analyses characterise a circuit around a large-signal periodic operating point (a mixer under its LO, a switched-capacitor under its clock, an oscillator). `.pss` finds that periodic steady state; the rest are small-signal analyses *on top of it*. All share the same leading arguments `<fund>` `<tstab>` `<outnode>` `<points>` `<harmonics>`

A self-contained example — a behavioural (B-source) conductance pumped by a 1 MHz "LO", which acts as a mixer:

```
* L0-pumped behavioural mixer
V1 lo 0 SIN(0 1 1meg)
VDC a 0 DC 1 AC 1
R1 a b 10k
B1 b 0 I=(1m + 0.8m*v(lo))*v(b) ; conductance modulated by the L0
C1 b 0 100p
.pss 1meg 1u b 1024 8 50 5u ; find the periodic steady state
.control
run
setplot pss2 ; frequency-domain result
plot mag(b)
.endc
```

```
.end
```

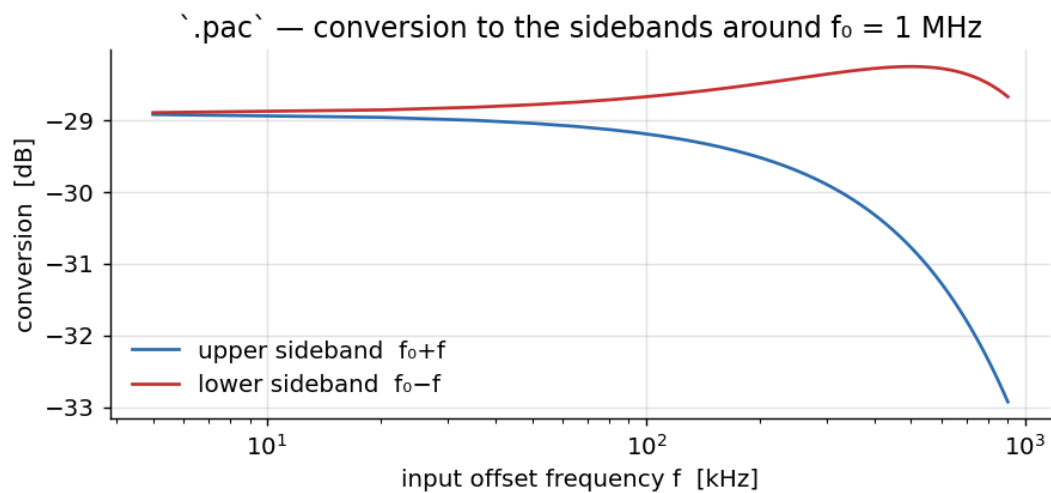
`.pss` returns the periodic operating point — its harmonic content shows the LO and the harmonics the nonlinearity generates:



With that operating point, `.pac` computes the small-signal conversion to each mixing sideband ($f_0 \pm f$) versus the input frequency — swap the `.pss` line for:

```
.pac 1meg 1u b 1024 6 50 5u lin 60 5k 900k 1
```

```
plot db(b_usb1) db(b_lsb1) ; upper / lower sideband conversion
```

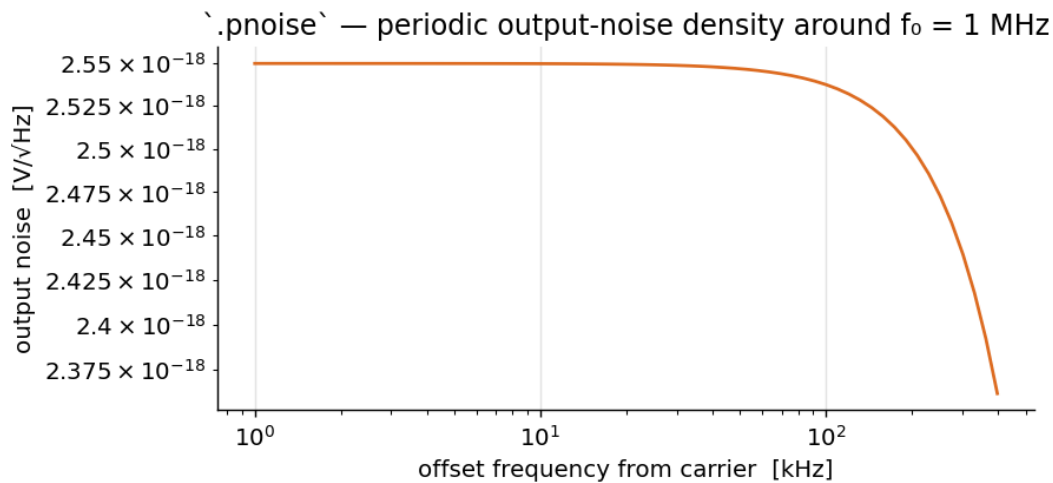


`.pnoise` gives the periodic / phase-noise density folded from every sideband, versus offset from the carrier:

```
.pnoise 1meg 1u b 1024 6 50 5u b vdc dec 25 1k 400k
```

```
set sqrnoise
```

```
plot onoise_spectrum ; V/√Hz vs offset frequency
```



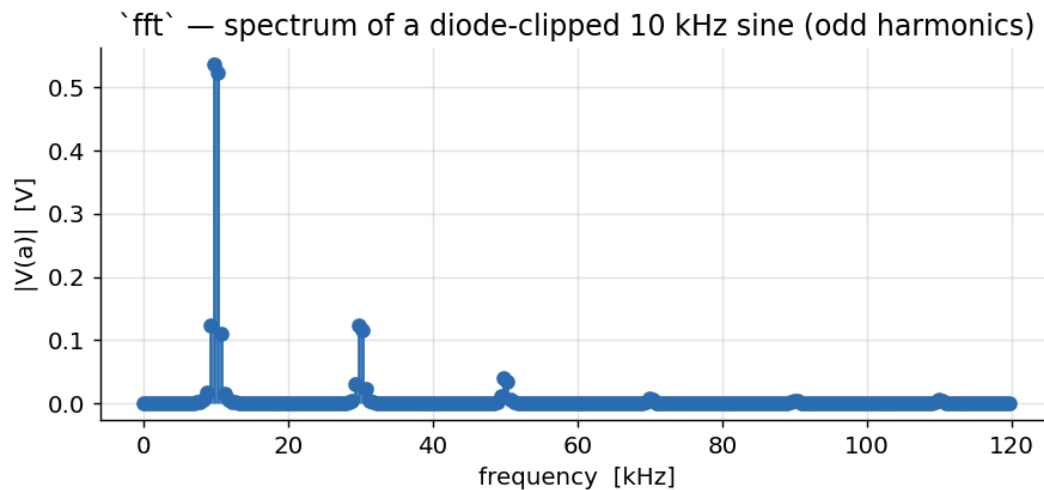
`.pxf` (periodic transfer function — sensitivity of an output to a small perturbation at each sideband) and `.psp` (periodic S-parameters) take the same `.pss` preamble; see the [RF suite reference](#) for those plus the two-tone `qpss/qpac/qnoise/qpxf` family and the oscillator `hbosc/phasenoise` flow.

4. Signal processing and measurement

These operate on vectors already in memory (typically after a `.tran`).

`fft <vec>` computes an FFT (run `linearize` first so the samples are uniformly spaced); `spec` and `psd` are related spectral views.

```
* harmonics of a diode-clipped 10 kHz sine
Vin in 0 sin(0 1.2 10k)
R1 in a 1k
D1 a 0 DL
D2 0 a DL
.model DL D(is=1e-12)
.tran 1u 2m
.control
run
linearize
fft v(a)
plot mag(v(a))
.endc
.end
```



The symmetric clipper produces only odd harmonics (30, 50, 70 kHz ...).

Command	Purpose
meas	measure a quantity from a result (meas tran tr TRIG ... TARG ..., meas ac gain MAX vdb(out), gain/phase margin, delays, RMS, ...)
fft	FFT spectrum of a vector
spec	spectrum over a frequency window: spec <fstart> <fstop> <fstep> <vec>
psd	power spectral density via FFT
fourier	Fourier coefficients + THD of a transient
eye	eye diagram + jitter: eye <expr> -ui <T> [-tstart t0] [-threshold v]
linearize	resample a transient onto a uniform time grid (needed before fft)
cutout	keep only a slice of a vector
transpose	transpose a multi-dimensional vector
cross	build a vector by picking elements from others
compose	build a vector from values/ranges: compose v start=0 stop=1 step=0.1
reshape	change a vector's dimensions
settype / deftype	set/redefine a vector's type (voltage, time, ...) for axis labelling

5. Vectors, variables and expressions

Results are vectors; `let` creates or modifies them and `print/plot` read them. Expressions support the usual math plus `v()`, `i()`, `vdb()`, `vp()`, `mag()`, `ph()`, `real()`, `imag()`, `db()`, `abs()`, `sqrt()`, `vector()`, `length()`, `mean()`, `vecmax()`, `vecmin()`, and the ternary `?:`.

```
.control
run
let pwr = v(out)*i(vout)           ; a new vector
let gain_db = vdb(out)
print vecmax(abs(v(out)))          ; scalars are 1-length vectors
print mean(pwr)
.endc
```

Command	Purpose
let	assign a vector: let x = expr

Command	Purpose
unlet	delete vectors
print	print vector values: print [col] expr ...
display	list the vectors in the current plot
define / undefine	user function: define sq(x) x*x
set / unset	set/clear a control variable: set numdgt=8
setcs	like set, but keeps the value's case
option / options	set a simulator option: option reltol=1e-4 klu
setscale	change the plot's default scale vector
setseed	reseed the RNG for reproducible Monte Carlo
remzerovec	drop zero-length vectors

6. Plotting and data export

```
.control
run
plot v(out) vs time
plot vdb(out) xlog          ; log-x Bode
pyplot acmag vdb(out)       ; matplotlib PNG/-window
wrdata out.dat v(out) v(in)  ; columnar (whitespace) data file
wrdata -csv out.csv v(out) v(in) ; comma-separated, with a header row
write out.raw v(out)         ; binary rawfile (reload with `load`)
.endc
```

Command	Purpose
plot	the built-in plotter: plot expr ... [vs expr] [xl lo hi] [yl lo hi] [xlog] [ylog]
pyplot	render/export via matplotlib (-eye, -hist, -contour variants); PNG when a file is named
gnuplot	render/export via gnuplot
hardcopy	write a PostScript/SVG hardcopy of a plot
asciiplot	plot as ASCII art in the terminal
wrdata	write columnar (whitespace) data — each vector prefixed by its scale; -csv (or set wr_csv) writes a comma-separated file with a header row and one shared scale column
write	write a binary rawfile (load reads it back)
load	load a rawfile into a new plot
wrsnp / rdsnp / wrs2p	Touchstone .sNp write / read / 2-port write
diff	compare two plots vector-by-vector
setplot	switch the current working plot
destroy	free a plot's data

7. Circuit and device management

```
.control
source amp.cir          ; read a deck
```

```

listing          ; show the parsed circuit
alter r1 = 2k     ; change a device value, then re-run
altermod nmos vth0=0.4 ; change a .model parameter
alterparam vsup=3.3 ; change a .param, then `reset` to apply
show m1          ; print device operating-point parameters
run
.endc

```

Command	Purpose
source	read and load a netlist file
listing	print the current circuit (logical/physical/deck)
edit	edit the deck in \$EDITOR, then reload
cirbyline	feed the circuit one line at a time (scripting)
alter / altermod / alterparam	change device / .model / .param values
show / showmod	print device / model parameters (operating point)
devhelp	list device types and their parameters
inventory	count the devices in the circuit
save	select which node/branch outputs to keep
setcirc	switch between loaded circuits
remcirc / removecirc	drop the current circuit
reset	rebuild the circuit (apply alterparam, clear state)
dump	dump the circuit matrix structure

8. Compiling and loading models

This build compiles Verilog-A to OSDI with openvaf-r and loads it directly.

```

.control
pre_osdi bsim4.osdi      ; load a compiled Verilog-A model (the `osdi` command)
pre_osdi -f bsim4.osdi   ; force-reload after recompiling (no restart)
pre_snp filter.s2p       ; compile a Touchstone file to an OSDI n-port, then load it
codemodel extra.cm       ; load an XSPICE code-model library
.endc
N1 d g s b bsim4mod      ; an OSDI device instance (N-prefix)
.model bsim4mod bsim4va w=10u l=1u

```

Command	Purpose
osdi (pre_ osdi)	load one/more compiled .osdi models; -f force-reloads a recompiled file
snp (pre_snp)	compile a Touchstone .sNp to a Verilog-A n-port OSDI model, then load it
codemodel	load an XSPICE code-model (.cm) library
use	load a device library

9. Statistics, yield and optimization

* Monte-Carlo yield of an RC corner

```

.param rr=agauss(1k, 5p, 3) ; 5% (3-sigma) Gaussian on R
R1 in out {rr}

```

```

C1 out 0 1n
Vin in 0 ac 1
.ac dec 20 1 1meg
.control
montecarlo 500 -analysis ac -spec 'vdb(out)' -min -3.5 -max -2.5
.endc
.end

```

Command	Purpose
montecarlo	packaged yield: N samples, per-spec pass/fail, yield + 95% CI
mcsample	Latin-Hypercube sampling of the .param random draws (lower variance)
highsigma	rare-event / high-sigma failure probability by scaled-sigma importance sampling
mccorr	register a correlation matrix; draw with mvnorm() in .param
setseed	reproducible random draws
sweep	sweep any knob (device/model/.param) and record outputs into a plot
optimize	Nelder-Mead / Levenberg-Marquardt parameter optimizer against an objective

See the [statistics](#) and [optimizer](#) references for full treatments with plots.

10. Post-layout and reliability

Command	Purpose
reduce	reduce an R/C parasitic network (TICER) to an equivalent .subckt valid to fmax
aging	age aging-capable devices to a target lifetime (HCI/NBTI/TDDDB) for fresh-vs-aged runs
emir	power-grid IR-drop + electromigration current density + Black's-equation MTTF

11. Interactive debugging and control

```

.control
stop when v(out) > 0.9      ; breakpoint on a condition
trace v(out)                ; watch a node
run                          ; runs until the breakpoint
where                        ; last non-converging node/device
resume                      ; continue
savestate ckpt.st           ; checkpoint the transient state to disk
* ... later, even in a fresh process ...
loadstate ckpt.st           ; restore and continue the .tran
.endc

```

Command	Purpose
run	run the analysis from the deck's dot-cards (optionally to a rawfile)
resume	continue after a breakpoint/stop
stop	set a breakpoint (stop after N, stop when expr)
step	advance N timepoints
trace / iplot	watch / incrementally plot nodes during a run
status / delete	list / remove breakpoints and traces

Command	Purpose
savestate / loadstate	checkpoint / restore the full transient state (cross-process)
snsave / snload	save / load a solution snapshot
where	print the last non-converging node or device

12. Control-flow scripting (**.control**)

.control blocks are a small shell-like language. In addition to the commands above:

```
.control
run
* structured control
if v(out) > 0.5
  echo "high"
else
  echo "low"
end
* loops
foreach f 1k 10k 100k
  ac dec 10 $f $f
  print vdb(out)
end
let k = 0
dowhile k < 3
  echo iteration $k
  let k = k + 1
end
.endc
```

Construct	Purpose
if / else / end	conditional
while / dowhile / end	pre-/post-test loops
foreach var v1 v2 ... / end	iterate over a value list
repeat [N] / end	repeat N times (or forever)
break / continue	loop control
label / goto	jump within a block
shift	shift a list/argv variable left
echo	print text/\$variables
shell	run a host shell command
cd / getcwd	change / print working directory
strcmp / strstr / strslice	string compare / search / slice into \$var
fopen / fread / fclose	open / read / close a file into shell variables

13. Help and utility commands

Command	Purpose
help / tutorial / newhelp / oldhelp	documentation browser (help <command>)
history	recall previous commands (history [-r] [N])
alias / unalias	define / remove a command alias
synhl	preview interactive syntax highlighting of a line
version	print the version
sysinfo	print host CPU/memory summary
rusage	print resource usage (time, memory, all)
cdump	dump the parsed control structures
mdump / mrdump	dump the matrix / RHS (debugging)
bug	open a bug report
rehash	rebuild the host-command hash
quit	exit ngspice

Developer/edge commands: aspice/rspice/jobs (async/remote runs), mc_source, optran, wrnodev, check_ifparm, bltplot (a plot alias), state (unimplemented placeholder).

Generated for the ngspice-46 + OpenVAF-r enhancement project. Plots produced by make_commands_figs.py from real ngspice runs; rebuild the PDF with build_commands_pdf.py.